AI Job Market Analysis

Project submitted to the

APSSDC

Bachelor of Technology

In

Computer Science and Engineering

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ABSTRACT:

This project aims to explore the current and future job opportunities in Artificial Intelligence (AI) and Machine Learning (ML) for the year 2025.

The main objective is to analyze which field AI or ML offers better career prospects, what skills are most important to get hired, and how salaries vary across countries and job roles. Data was collected from global job portals, industry reports, and salary databases.

Using Python, along with libraries such as Pandas, Matplotlib, and Seaborn, the project performs data cleaning, visualization, and trend analysis. The results highlight that skills like LLMs (Large Language Models), MLOps, Deep Learning, and NLP are in high demand. ML Engineers and AI Specialists are among the highest-paid professionals.

This study provides clear insights for students, professionals, and job seekers to focus on the right skill sets and prepare for the evolving AI/ML job market in 2025. The project can be extended in the future to forecast job and salary trends beyond 2025 using predictive models.

INTRODUCTION:

This project focuses on analyzing a large dataset of over 15,000 job postings collected from various global sources. The dataset contains rich information including job titles, salaries (in USD and local currencies), experience levels, employment types, company size and location, required skills, education levels, and more.

The primary objectives of this analysis are:

- To identify the highest-paying job roles and industries.
- To examine the influence of experience level, education, and company size on salaries.
- To analyze the demand for specific technical and soft skills.
- To understand the prevalence and impact of remote work opportunities.
- To extract patterns and insights that can guide job seekers, HR professionals, and policymakers.

SYSTEM REQUIREMENTS:

SOFTWARE REQUIREMENT:

- **OPERATING SYSTEM:**

  Windows/Linux

- **Python:**

  Python 3.x is required for running the analysis. Make sure you have the latest stable version of Python installed.

- **Libraries:**

  Pandas: Install the Pandas library using pip, a package manager for Python.

  ```
  pip install pandas
  ```

Matplotlib: Install the Matplotlib library using pip

pip install matplotlib

Seaborn: Install the seaborn library using pip command

pip install seaborn

- **Jupyter Notebook**

HARDWARE REQUIREMNTS:

- Standard PC with sufficient RAM and disk space

ARCHITECTURE:

The architecture of the AI Job market analysis using Python, Matplotlib, and Pandas involves several key steps that form a cohesive workflow. The process typically includes data overview, data preprocessing, exploratory data analysis (EDA), data visualization.

Let's explore the architecture in more detail:

Data overview:

Importing libararies:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

printing first 5 rows:

```
df.head()
```

	job_id	job_title	salary_usd	salary_currency	experience_level	employment_type	company_location	company_size	employee_residence	remote_ratio
0	AI00001	AI Research Scientist	90376	USD	SE	CT	China	M	China	50
1	AI00002	AI Software Engineer	61895	USD	EN	CT	Canada	M	Ireland	100
2	AI00003	AI Specialist	152626	USD	MI	FL	Switzerland	L	South Korea	0
3	AI00004	NLP Engineer	80215	USD	SE	FL	India	M	India	50
4	AI00005	AI Consultant	54624	EUR	EN	PT	France	S	Singapore	100

basic statistics of dataset:

```
#display basic statistics
df.describe()
```

	salary_usd	remote_ratio	years_experience	job_description_length	benefits_score
count	15000.000000	15000.000000	15000.000000	15000.000000	15000.000000
mean	115348.965133	49.483333	6.253200	1503.314733	7.504273
std	60260.940438	40.812712	5.545768	576.127083	1.450870
min	32519.000000	0.000000	0.000000	500.000000	5.000000
25%	70179.750000	0.000000	2.000000	1003.750000	6.200000
50%	99705.000000	50.000000	5.000000	1512.000000	7.500000
75%	146408.500000	100.000000	10.000000	2000.000000	8.800000
max	399095.000000	100.000000	19.000000	2499.000000	10.000000

Data Preprocessing:

Data preprocessing is a crucial step to ensure data quality and consistency.

Using Python's Pandas library, the data is loaded into a DataFrame, allowing for easy data manipulation and analysis.

This step involves handling missing values, handling duplicates, converting data types, and addressing any data quality issues.

Checking for missing values:

```
df.isnull().sum()
```

```
job_id                    0
job_title                 0
salary_usd                0
salary_currency           0
experience_level          0
employment_type           0
company_location          0
company_size              0
employee_residence        0
remote_ratio              0
required_skills           0
education_required        0
years_experience          0
industry                  0
posting_date              0
application_deadline      0
job_description_length    0
benefits_score            0
company_name              0
dtype: int64
```

there are no missing values in the data

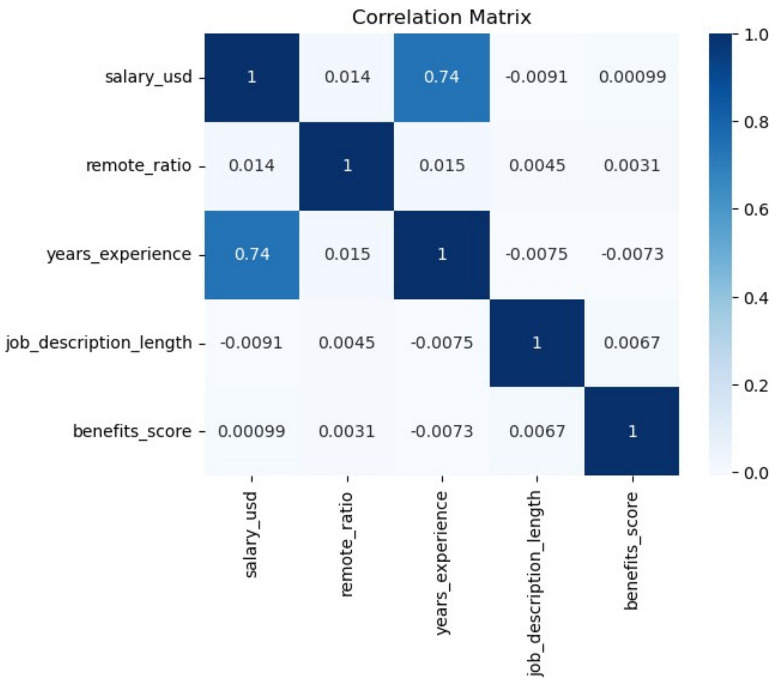
Exploratory Data Analysis (EDA):

EDA involves exploring the data to gain insights into its structure, distributions, and relationships between variables.

Pandas functions are used to perform summary statistics, groupings, and aggregations to understand key trends and patterns.

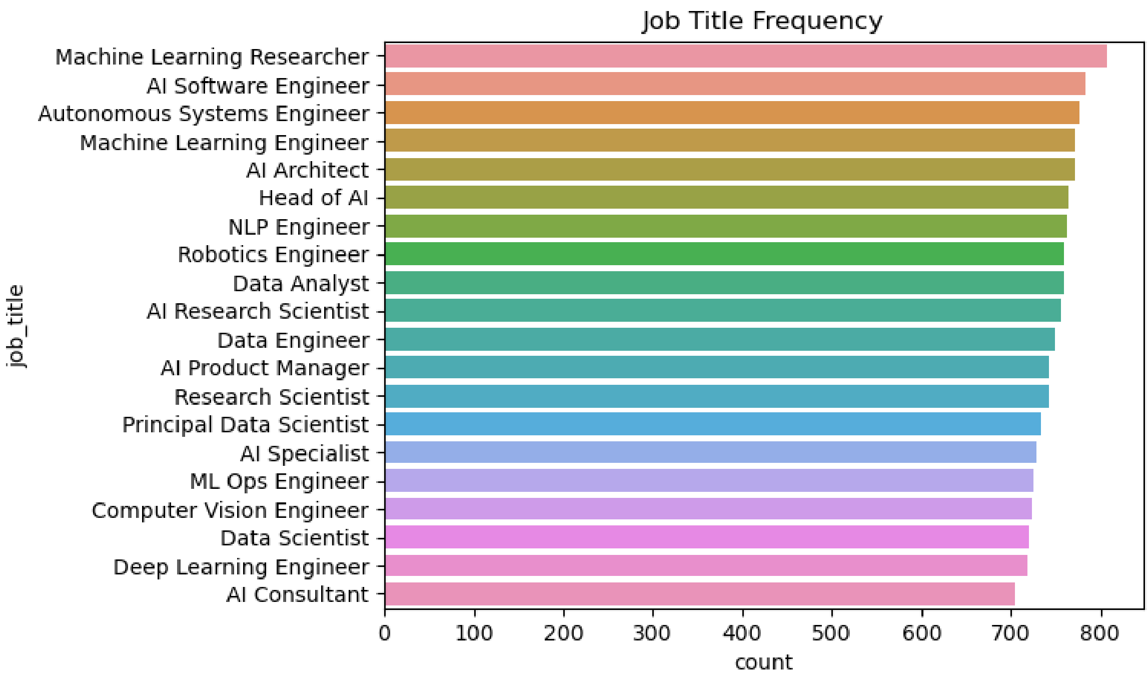
Data visualization with Matplotlib helps to create plots, histograms, scatter plots,

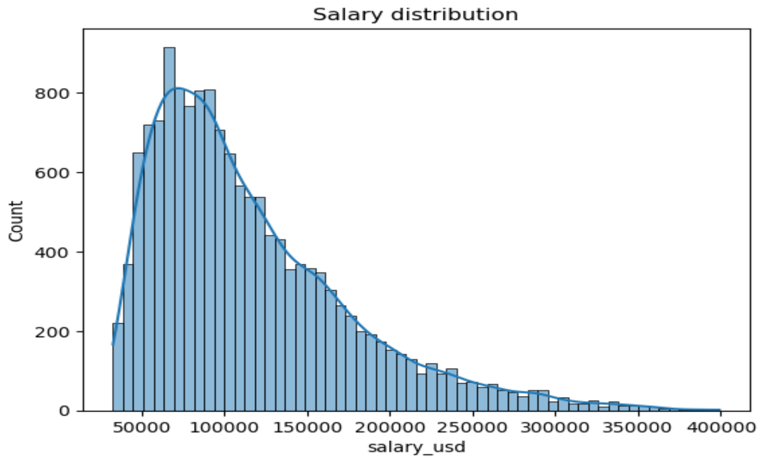
and other visualizations to visualize patterns and correlations effectively.



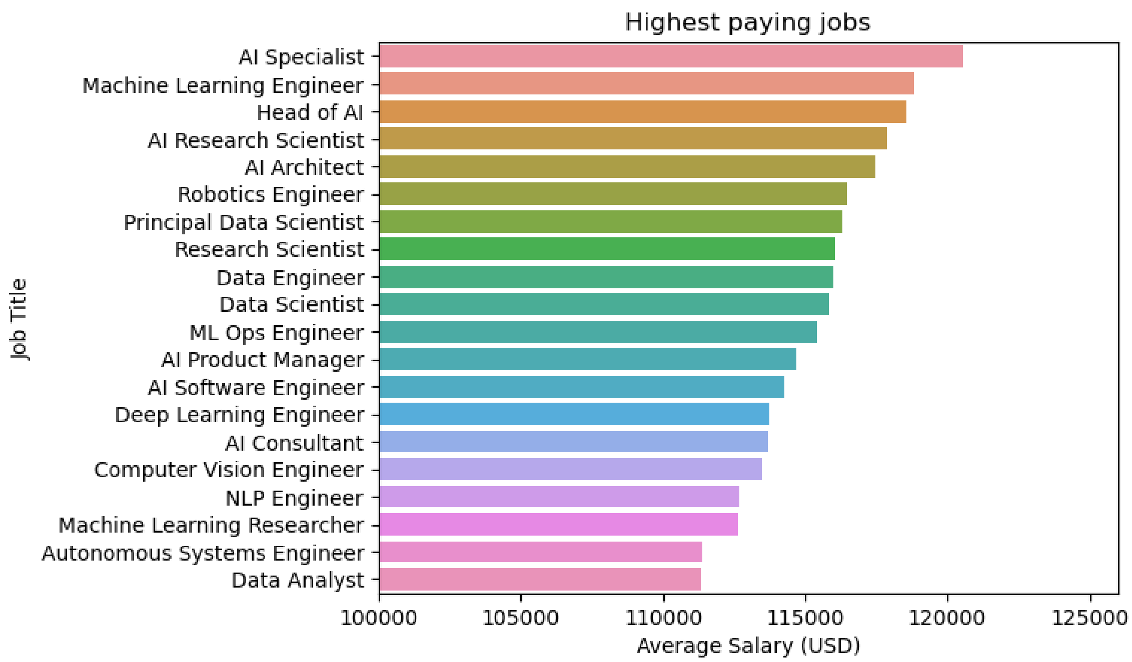
from this heat map we can see that years_experience and salary_usd columns are highly correlated.

From this line plot we can observe that salary is increasing over years of experience.

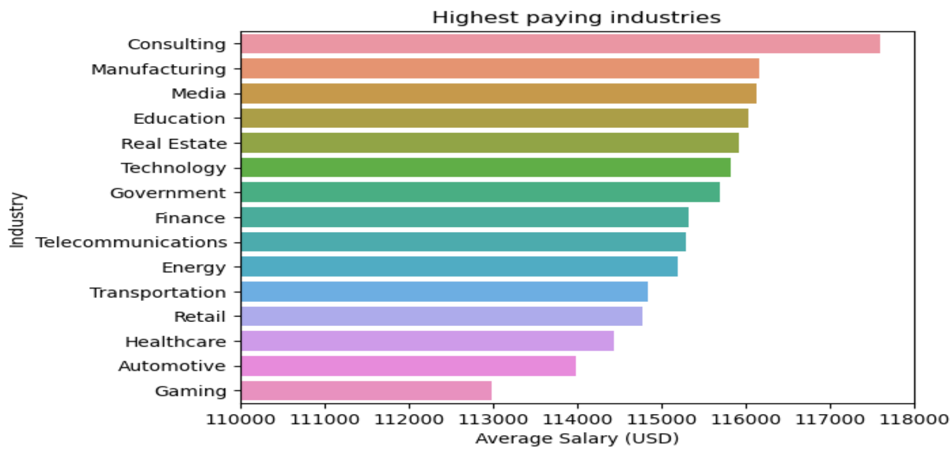




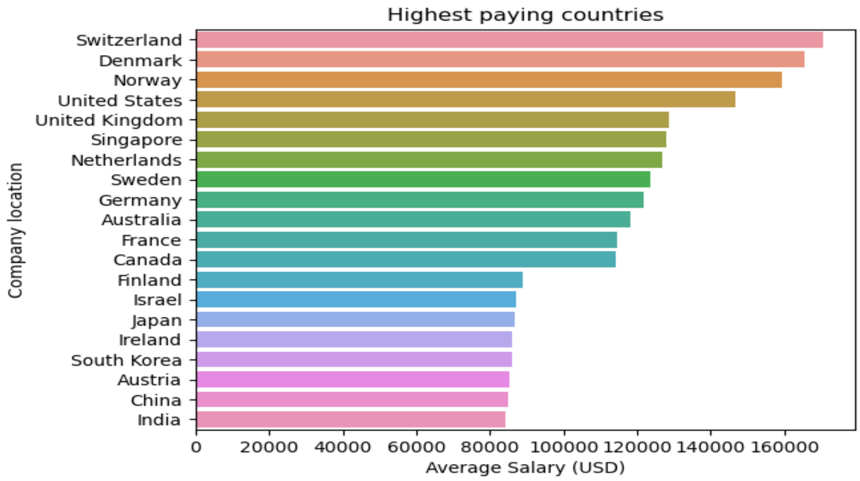
From this plot, most of the jobs are in the salary range of 75000 to 80000.



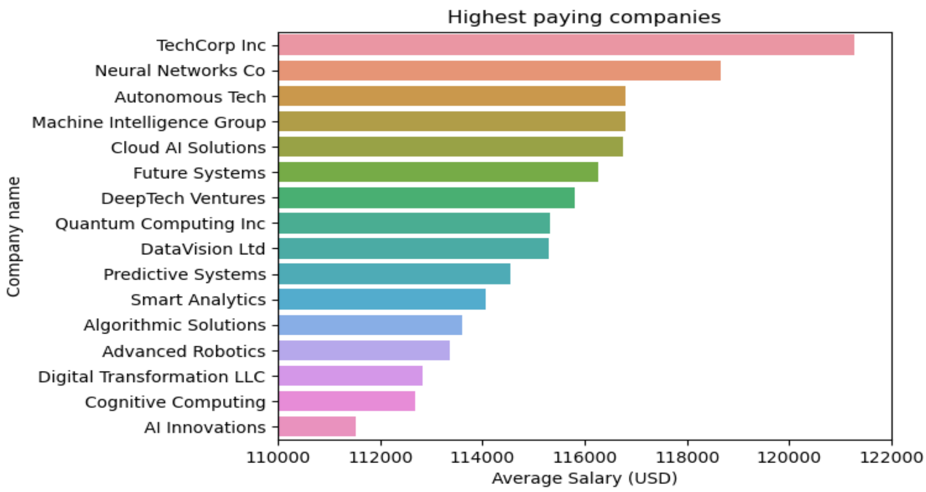
from this plot, highest paying jobs are AI Specialist, ML engineer, Head of AI.



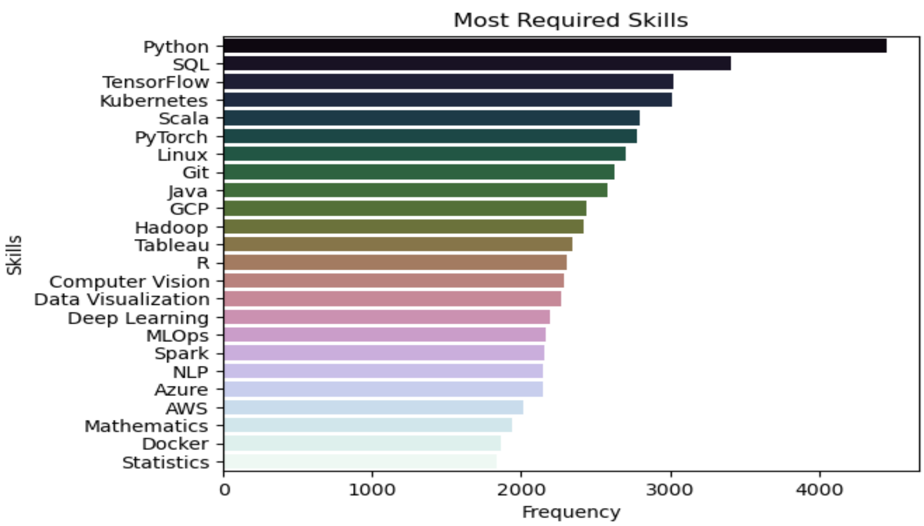
Consulting, Manufacturing, Media are the highest paying industries.



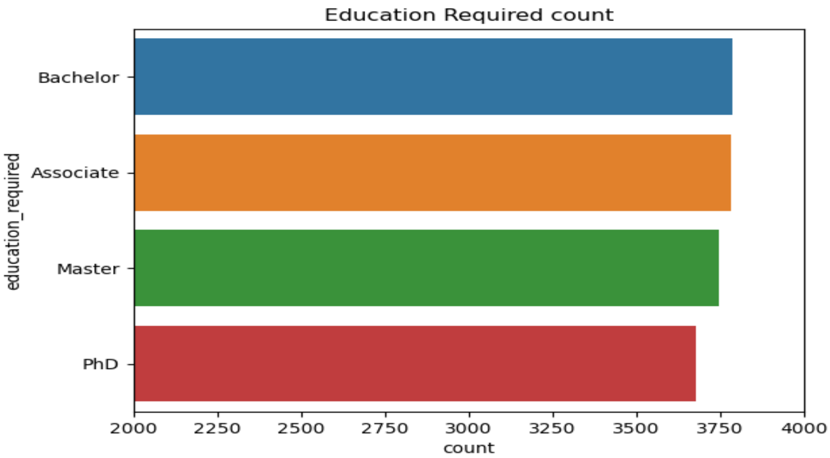
Companies in Switzerland, Denmark, Norway are paying more salary.



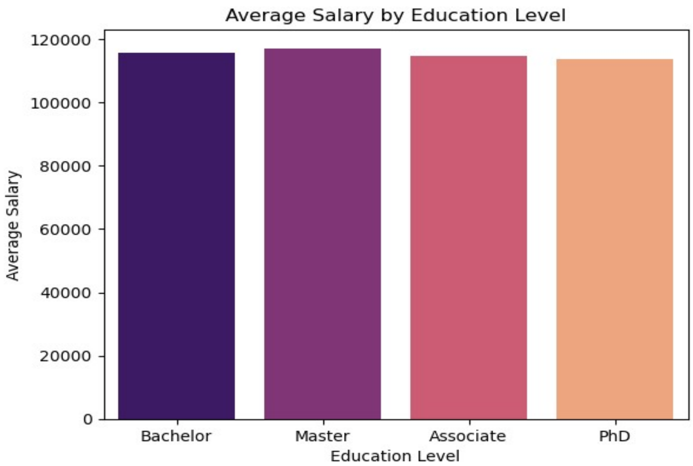
TechCorp Inc, Neural Networks Co, Autonomous Tech are highest paying companies.



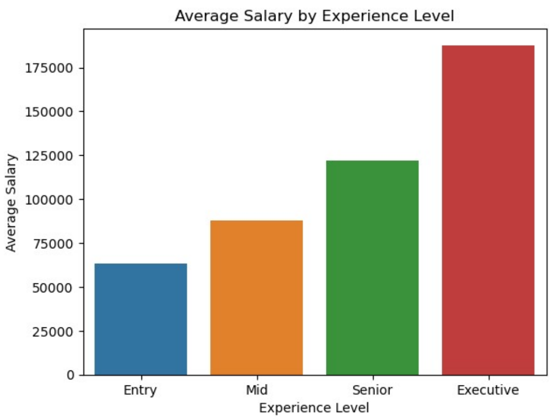
From this plot we can see that Python is the most required skill since it is used mostly for ml and ai related works. Python, SQL, Tensorflow are top required skills.



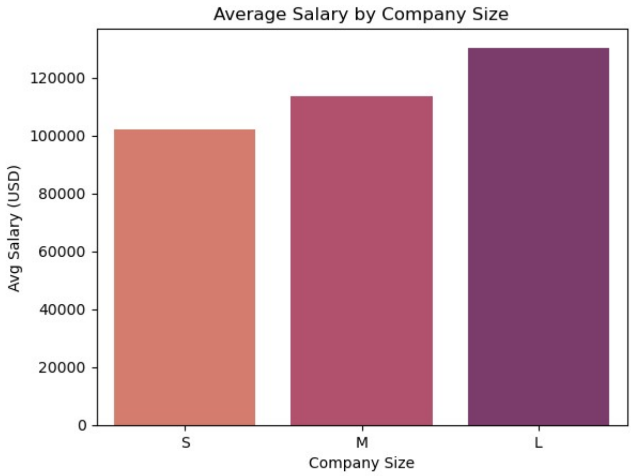
Most of the job roles require atleast Bachelor degree.



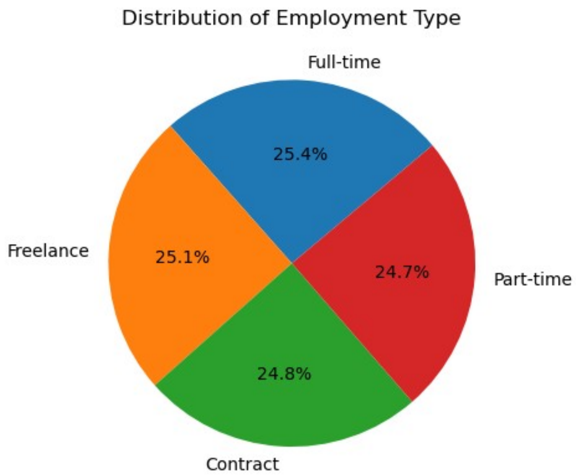
There is not much difference in average salary eventhough education is high.



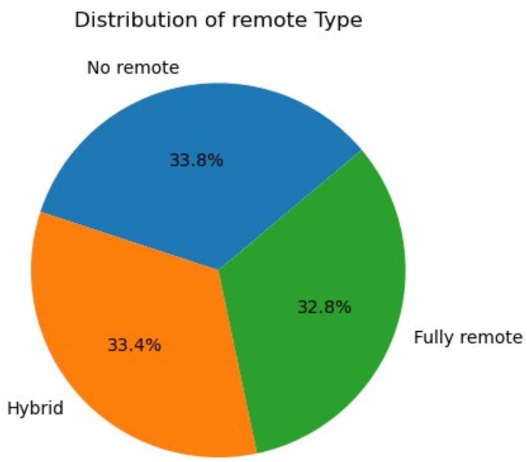
From these plots we can observe that salary is high for more experience level.



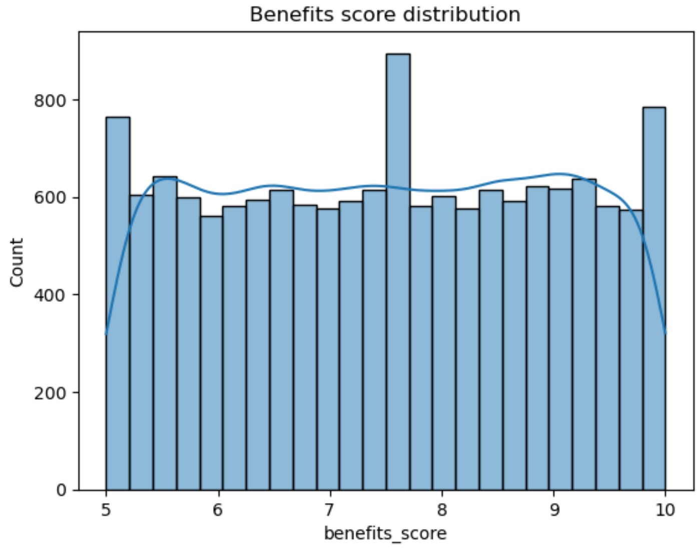
Salary is high in companies with large size.



All employment types are almost equally distributed.



All remote types are equally distributed.



Many jobs have benefits score between 7 to 8.

ADVANTAGES:

1. Informed Career Planning

Helps job seekers and students understand which roles, skills, and industries offer the best salary and growth potential, enabling smarter career decisions.

2. Skill Demand Awareness

Identifies the most frequently required skills (e.g., Python, Machine Learning), guiding professionals on where to focus their learning and upskilling.

3. Understanding Remote Work Trends

Offers valuable insights into the availability and pay structure of remote, hybrid, and on-site jobs, which is crucial in today's flexible work environment.

4. Salary Benchmarking

Enables comparison of salary trends across different roles, locations, and experience levels, helping companies remain competitive in their compensation strategies.

5. Employer Insights

Provides recruiters and HR departments with data-driven evidence on what top talent looks for — from benefits to flexibility — to enhance their job offerings.

6. Industry Trend Monitoring

Identifies which sectors are actively hiring and investing in talent, helping policymakers, educators, and institutions align their training programs accordingly.

7. Global Market View

Since the dataset includes global postings, it helps highlight location-specific trends — useful for remote job seekers or global hiring initiatives.

8. Efficient Job Matching

By analyzing roles and skill requirements, platforms and individuals can better match candidates with the right job openings.

CONCLUSION:

This analysis of over 15,000 job postings provides valuable insights into the current trends shaping the global job market. By exploring salary distributions, experience requirements, remote work availability, in-demand skills, and industry trends, the report helps illuminate the factors that define high-value job opportunities today.

The findings reveal that technical roles particularly in AI, machine learning offer some of the highest salaries. These roles are increasingly remote-friendly and demand a strong foundation in key skills such as Python, SQL, and tensorflow. Furthermore, senior-level experience and employment at large companies significantly increase salary potential.

For individuals, the insights serve as a practical guide to career planning, highlighting which skills to develop and what roles to target.

Overall, the project demonstrates the power of data analysis in uncovering meaningful patterns in large datasets and supports informed decision-making in both professional development and strategic hiring.

REFERENCES:

- Global AI Job Market & Salary Trends 2025 : https://www.kaggle.com/datasets/bismasajjad/global-ai-job-market-andsalary-trends-2025
- Seaborn documentation :

  https://seaborn.pydata.org/tutorial.html
- Matplotlib.pyplot documentation :

  https://matplotlib.org/stable/api/pyplot_summary.html